

Walmart Sales Analysis

A comprehensive SQL-based data analysis project examining sales performance, customer behavior, and operational efficiency across Walmart stores

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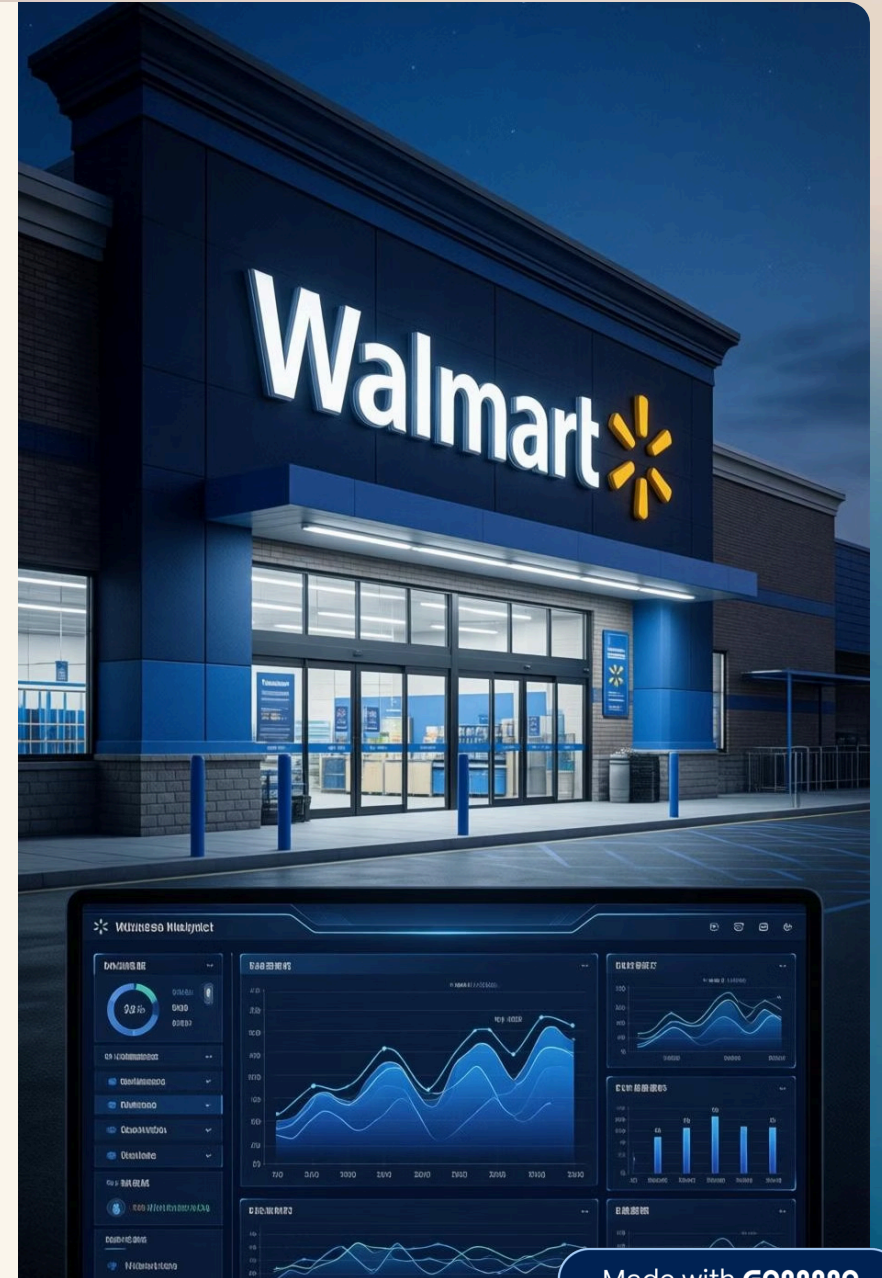


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Executive Summary

1

Food & Beverages is the top revenue driver (\$53,471 in sales)

2

Evening hours generate maximum sales (279 transactions after 6 PM)

3

Naypyitaw is the highest-performing city (\$5,265 gross income)

4

Member customers contribute more revenue & VAT than normal customers

5

Customer satisfaction is highest in mornings (peak rating periods)

Transforming Retail Data Into Strategic Insights

Business Objective

Leverage SQL analytics to uncover sales patterns, optimize inventory management, and enhance customer satisfaction across Walmart's retail network.

Tools & Technologies

- Microsoft SQL Server
- Advanced SQL queries & aggregations
- Data-driven business intelligence





DATASET

Understanding the Data Foundation

Transaction Records

1,000 sales transactions across three major cities: Yangon, Mandalay, and Naypyitaw

Time Period

January through March 2019, capturing quarterly sales performance and seasonal trends

Key Metrics

17 data points per transaction including sales, ratings, payment methods, and customer demographics

Critical Business Questions

This analysis addresses five strategic priorities for Walmart's operational excellence:

1

Product Performance

Which product lines generate the highest revenue and customer engagement?

2

Geographic Insights

How do sales vary across branches and cities?

3

Customer Behavior

What patterns emerge from member vs. normal customer purchasing habits?

4

Temporal Trends

When do peak sales occur by time of day, day of week, and month?

5

Payment Preferences

How do payment methods correlate with transaction values and customer types?

Strategic Business Questions (Continued)

1

Customer Segmentation

How can we segment customers to tailor marketing strategies and product offerings more effectively?

2

Profitability Analysis

Which product categories or branches yield the highest profit margins after accounting for operational costs?

3

Inventory Turnover

How efficiently is inventory moving, and are there opportunities to optimize stock levels to reduce carrying costs or prevent stockouts?

4

Market Penetration

What is our current market penetration in each city, and what strategies can increase our customer base?

5

Competitive Positioning

How do our pricing, product variety, and customer service compare to key competitors, and where can we gain an advantage?

Advanced Query Techniques

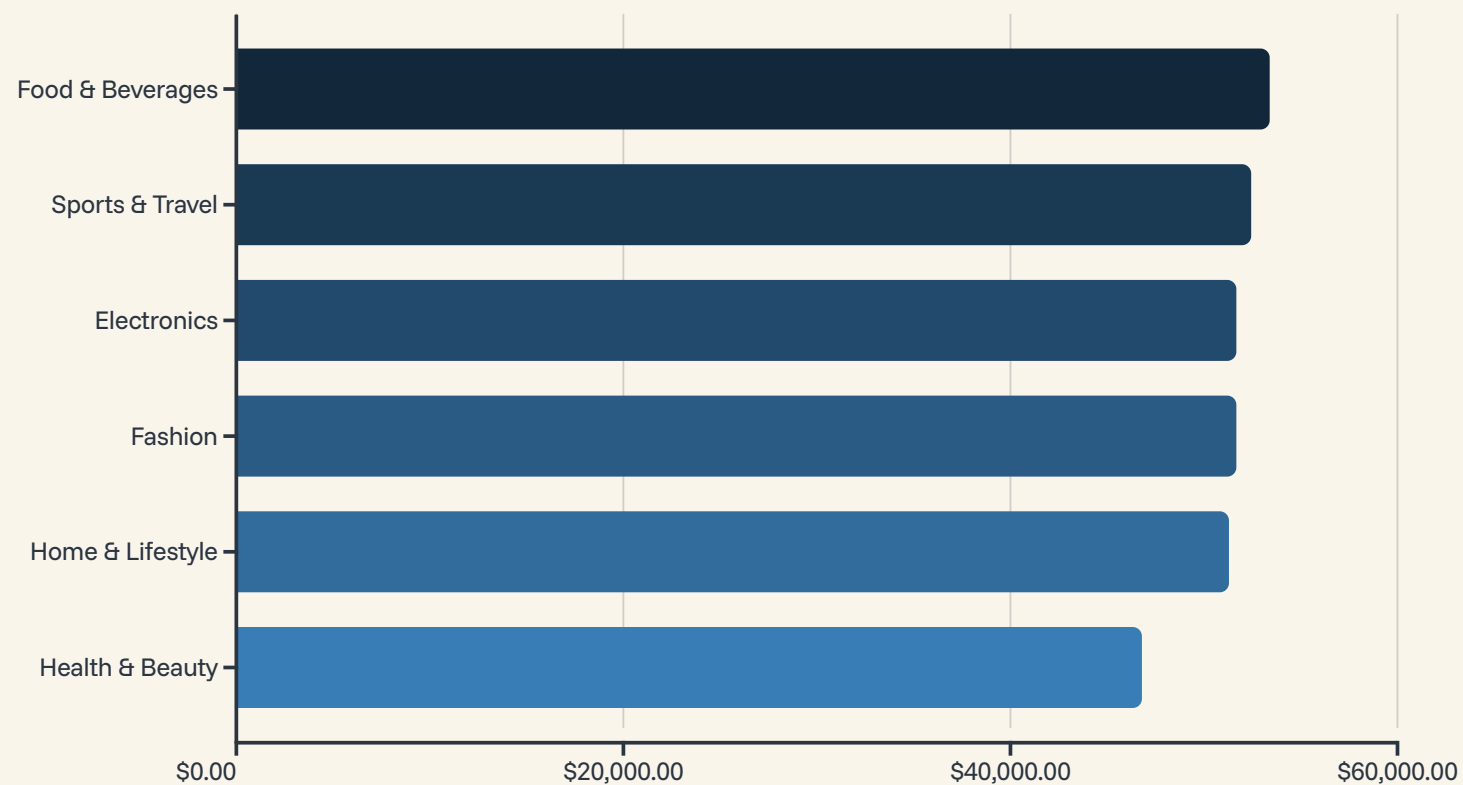


Technical Approach

Implemented 15+ complex SQL queries leveraging advanced techniques:

- **Aggregate functions** for sales totals and averages
- **Window functions** for cumulative income analysis
- **Date functions** for temporal pattern identification
- **Subqueries** for comparative benchmarking
- **GROUP BY & JOIN** operations for multi-dimensional analysis
- **Common Table Expressions (CTEs)** for break complex logic into readable steps

Product Line Performance Analysis



Food and Beverages leads with \$53,471 in sales, while Sports & Travel and Electronics maintain competitive positions. Health & Beauty presents growth opportunity with lowest revenue at \$46,851.



Key Performance Insights

1

Total Revenue

Combined sales across all branches and product lines during Q1 2019

2

Average Rating

Customer satisfaction score with Branch C leading at 7.07 stars

3

Top Category Volume

Fashion Accessories sold the highest quantity of units

4

Highest City Income

Naypyitaw generated the most gross income among three cities

TRENDS

Temporal & Behavioral Patterns

Monthly Sales Trajectory

January peaked at \$110,754 in sales, followed by moderate decline through March (\$104,243). February showed strongest gross income efficiency.

Peak Shopping Hours

279 transactions occurred after 6 PM, with March recording the highest evening activity (105 sales), indicating strong after-work shopping patterns.

Weekend Sales Activity

Significant weekend transaction volume demonstrates consistent customer engagement on Saturdays and Sundays across all product categories.

Payment Method Distribution

Cash, credit card, and e-wallet payments show balanced adoption, with cash maintaining a strong preference for high-value purchases exceeding \$900.



Strategic Business Recommendations



Inventory Optimization

Increase stock levels for Food & Beverages and Sports & Travel. Consider strategic promotions for Health & Beauty to boost performance.



Targeted Marketing

Launch evening promotional campaigns (6-9 PM) and weekend-specific offers to capitalize on peak traffic periods.



Branch Strategy

Replicate Branch C's success factors (7.07 rating) across other locations. Invest in Naypyitaw's high-performing market presence.



Customer Engagement

Develop membership incentive programs to convert Normal customers. Analyze quantity purchase patterns for bulk-buying promotions.

Project Impact & Next Steps

Key Achievements

- Identified \$53K revenue opportunity in top product line
- Discovered 279 high-value evening transactions
- Mapped customer satisfaction patterns by branch
- Revealed seasonal sales trends for forecasting

Future Enhancements

1. Build interactive Power BI dashboard
2. Implement predictive analytics models
3. Expand analysis to multi-year trends
4. Integrate real-time sales monitoring



Connect with me: Manohar K | Data Analyst | *Transforming data into actionable retail intelligence*